

RICHMOND, VA, USA

WMO: 724010

Lat: 37.512N

Lon: 77.323W

Elev: 50

StdP: 100.73

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.0	-18.4	0.7	-4.9	-15.8	0.9	-2.9	11.3	8.2	10.3	6.8	3.1	0	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.9	24.2	33.6	23.9	32.3	23.4	25.8	31.7	25.2	30.9	24.6	29.8	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	19.3	28.2	23.7	18.6	27.5	23.1	18.0	26.9	79.8	31.9	76.9	30.8	74.6	29.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.4	7.5	-12.9	37.1	3.2	1.7	-15.2	38.4	-17.1	39.4	-18.9	40.4	-21.2	41.6		
			-13.6	27.0	3.0	0.6	-15.8	27.5	-17.5	27.9	-19.2	28.2	-21.4	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	3.6	5.2	9.3	15.0	19.6	24.2	26.3	25.5	22.0	15.9	9.9	5.8
	DBStd	9.10	5.93	5.36	5.38	4.75	4.19	3.25	2.54	2.52	3.34	4.41	4.77	5.10
	HDD10.0	668	213	151	80	9	0	0	0	0	0	4	59	151
	HDD18.3	2019	457	367	283	121	38	2	0	0	8	102	253	388
	CDD10.0	2586	16	18	59	161	297	426	505	479	360	187	58	21
	CDD18.3	894	0	0	3	23	76	178	247	221	118	26	2	1
	CDH23.3	8292	1	3	41	250	681	1697	2489	2038	879	199	10	3
	CDH26.7	3324	0	0	6	71	222	702	1133	854	290	47	1	0
Wind	WSAvg	3.3	3.6	3.7	4.0	4.0	3.4	3.2	2.9	2.7	2.9	3.0	3.2	3.2
Precipitation	PrecAvg	1118	81	73	97	77	99	104	121	116	101	86	78	84
	PrecMax	1619	202	152	220	211	263	338	353	415	422	238	244	207
	PrecMin	718	16	12	5	16	22	10	13	14	2	0	4	10
	PrecStd	215	39	34	47	38	52	65	66	71	81	56	50	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	23.2	27.4	31.3	33.0	36.1	37.3	36.0	33.9	30.7	24.8	22.8
		MCWB	16.2	16.3	16.8	19.6	22.3	24.0	24.8	24.7	23.1	21.6	18.0	16.7
	2%	DB	18.7	19.9	24.1	28.8	31.0	33.9	35.1	34.3	31.8	27.7	22.4	19.4
		MCWB	15.1	13.9	15.4	18.2	21.8	23.7	24.6	24.1	22.6	20.5	16.7	16.2
	5%	DB	16.1	17.0	21.3	26.4	29.3	32.3	33.7	32.7	29.9	25.5	20.3	16.8
		MCWB	12.8	12.0	13.9	17.3	21.1	23.1	24.3	23.8	21.8	19.4	15.7	13.7
	10%	DB	13.1	14.3	18.6	23.8	27.4	30.7	32.2	31.1	28.1	23.4	18.1	14.1
		MCWB	9.6	9.8	12.7	16.3	20.0	22.6	24.0	23.3	21.2	18.4	14.3	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	17.9	18.7	21.8	24.2	25.6	26.9	26.7	25.2	24.0	20.3	18.8
		MCDB	20.0	21.8	24.2	26.9	30.0	32.0	33.3	32.6	29.5	26.5	22.4	20.8
	2%	WB	15.8	15.5	17.1	20.2	23.0	24.8	26.0	25.7	24.4	22.3	18.5	16.6
		MCDB	18.3	18.3	22.0	25.9	28.7	31.0	32.2	31.4	28.5	25.6	21.0	19.0
	5%	WB	13.3	13.0	15.6	18.8	22.1	24.3	25.4	25.1	23.6	20.8	16.7	14.3
		MCDB	15.5	15.9	19.8	24.1	27.4	30.2	31.4	30.4	27.3	24.5	19.4	16.2
	10%	WB	10.1	10.3	13.7	17.6	21.1	23.7	24.8	24.4	22.7	19.3	15.0	11.5
		MCDB	12.5	13.8	17.7	22.4	25.8	29.0	30.2	29.0	26.1	22.6	17.6	13.5
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.7	11.5	12.3	11.2	10.6	10.3	10.2	10.2	11.1	11.0	9.7
		MCDBR	13.3	14.6	15.0	15.2	13.0	12.3	12.1	12.0	12.2	12.7	13.3	12.6
		MCWBR	10.2	10.0	8.4	7.0	5.8	4.5	3.9	3.9	4.7	6.5	8.2	9.5
	5% WB	MCDBR	12.2	13.1	13.2	12.9	11.3	10.9	10.7	10.0	9.3	10.5	10.6	11.2
		MCWBR	10.5	10.6	8.9	7.1	5.4	4.5	4.0	4.0	4.5	6.1	8.0	9.7
Clear-Sky Solar Irradiance	taub	0.301	0.314	0.346	0.390	0.427	0.486	0.505	0.506	0.407	0.353	0.320	0.304	
	taud	2.532	2.484	2.432	2.325	2.246	2.127	2.095	2.076	2.339	2.490	2.537	2.556	
	Ebn at Noon	893	921	918	891	859	805	786	775	842	865	862	864	
	Edh at Noon	77	91	106	125	138	155	159	159	115	90	77	71	
All-Sky Solar Radiation	RadAvg	2.31	3.07	4.07	5.12	5.82	6.32	6.02	5.44	4.43	3.47	2.61	2.04	
	RadStd	0.19	0.34	0.33	0.40	0.54	0.44	0.30	0.26	0.46	0.41	0.21	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+161	+107
Regional (72 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	+91

Nomenclature: See separate page